

IFUL: A Python package for joint-modeling strong gravitational lensing and their source kinematics

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Summary

IFUL (Integral Field Unit Lensing) is an open-source Python package for joint modeling and simulating strong gravitational lensing systems together with the spatially resolved internal kinematics of background source galaxy. Strong gravitational lensing occurs when the gravitational field of a massive foreground galaxy or galaxy cluster (the “lens”) bends light emitted by a background galaxy (the “source”), magnifying and distorting its appearance into multiple images or arcs. This phenomenon serves as a primary probe in observational astrophysics for measuring cosmological parameters (e.g., the Hubble constant H_0 , the dark energy equation of state w and the matter density of the universe $\Omega_{m,0}$), testing dark matter models, and studying high-redshift galaxy structure and evolution across time.

While traditional gravitational lens modeling tools rely primarily on two-dimensional imaging data, IFUL incorporates three-dimensional Integral Field Unit (IFU) spectroscopic datacubes into an end-to-end forward-modeling framework. Integral field spectroscopy captures two-dimensional spatial images where every pixel (spaxel) contains a full spectrum, providing spatially resolved measurements of the internal stellar motion of the galaxy. This allows for our framework to not only flux, but also dynamical information into our model. IFUL forward-models every spaxel in the 3D datacube by unifying macro lens mass profiles with physical models of source galaxy kinematics. By leveraging spatially resolved kinematic markers within lensed arcs and isolating source line emission from foreground lens light, IFUL provides tighter constraints on both lens mass distributions and high-redshift source dynamics than just using the imaging data alone.

Statement of need

Strong gravitational lens modeling has entered a regime where precision cosmography and mass profile determinations are primarily limited by systematic modeling uncertainties rather than statistical noise. Traditional lens modeling pipelines rely on two-dimensional photometric imaging. While three-dimensional Integral Field Unit (IFU) spectroscopy—from space and ground-based observatories such as *JWST* (NIRSpec, MIRI), VLT (MUSE, ERIS), Keck (KCWI/KCRM), and ALMA—is routinely acquired for lensing fields, these datacubes are typically underutilized. In conventional workflows, IFU data are restricted to measuring integrated deflector stellar velocity dispersions or confirming source galaxy redshifts, discarding the rich three-dimensional spatial-spectral information contained in individual spaxels.

IFUL makes several improvements to traditional 2D lens modeling:

- Kinematic markers as constraints: When a source galaxy exhibits coherent rotation or dynamic structure, velocity gradients serve as spatial markers across multiple lensed

images. Mapping these kinematic features in the image plane provides powerful complementary constraints on the lens mass profile.

- Continuum subtraction and image disambiguation: By performing continuum subtraction across the 3D datacube, IFUL isolates narrow emission lines belonging to the background source galaxy. This removes severe flux contamination from the bright foreground lens, unblending overlapping components and revealing faint or hidden counter-images that might otherwise remain buried under lens continuum light.
- Systematic uncertainties and degeneracies: As lens modeling becomes systematic-dominated, IFUL provides a novel, independent modeling avenue. Jointly modeling spatial flux and source velocity fields (v_{los} and σ_v) helps break classic mass profile degeneracies (such as radial slope and mass-sheet transformations) that cannot be resolved by 2D imaging alone.
- Understanding source galaxy dynamics: By forward-modeling the macro lens concurrently with the 3D IFU datacube, IFUL constructs intrinsic kinematic maps of the background galaxy that integrate spectroscopic information across all multiple lensed images. Combining spaxel data from every image into a unified source-plane model enables a robust determination of the source galaxy's intrinsic line-of-sight velocity (v_{los}) and velocity dispersion (σ_v) fields. This provides a detailed, high-resolution view of high-redshift galaxy kinematics and dynamics while fully marginalizing over lens-model parameters.

General improvements to lens mass modeling directly benefit diverse science goals, including time-delay cosmography (H_0), dark matter substructure searches, and multi-plane dark energy measurements.

Despite the growing availability of high-resolution IFU observations from current facilities and upcoming Extremely Large Telescope IFU instruments (such as ELT/HARMONI, ELT/MICADO, and TMT/IRIS), there is currently no existing open-source software dedicated to full 3D forward-modeling of lensed IFU datacubes. Individual research groups attempting 3D lens modeling must write custom, non-standardized codebases from scratch, posing a major barrier to entry and hindering scientific reproducibility. IFUL fills this software gap by delivering an open-source, modular, and user-friendly Python package built to interface with established lens modeling tools like *lenstronomy* (Birrer et al., 2021; Birrer & Amara, 2018).

IFUL is designed for astronomers, astrophysicists, and observational cosmologists working on strong gravitational lensing, high-redshift galaxy kinematics, and 3D spectroscopic analysis.

State of the field

Strong gravitational lens mass modeling is supported by a mature ecosystem of open-source software packages in astrophysics. The most widely adopted Python package is *lenstronomy* (Birrer et al., 2021; Birrer & Amara, 2018), which provides a flexible, multi-purpose framework for modeling two-dimensional (2D) imaging data, time-delay cosmography, and dark matter substructure. Other notable 2D lens modeling software packages include *PyAutoLens* (?), *glafic* (?), *gigalens* (?), *gravlens/lensmodel* (?), and *GLEE* (?; ?). While these frameworks excel at modeling high-resolution, multi-band photometric images (e.g., from the *Hubble Space Telescope* or *JWST* imaging filters), they operate exclusively on 2D spatial pixel arrays. Consequently, existing tools lack the architectural data structures and forward-modeling engines necessary to process three-dimensional (3D) Integral Field Unit (IFU) spectroscopic datacubes (x, y, λ), which combine spatial morphology with wavelength-resolved spectral line dynamics.

The scientific advantage of combining 3D IFU spectroscopy with strong gravitational lensing has been recognized in observational cosmology, yet no general-purpose, open-source software package has been released to serve this need:

- 88 ▪ Early 3D implementations: (?) developed an early 3D IFU lens modeling code in
89 IDL to analyze source velocity fields in strong lens systems. However, this code was
90 proprietary, never publicly released, and was not expanded into a general-purpose software
91 package. Moreover, IFU instrumentation, spatial resolution, and sensitivity have advanced
92 dramatically over the past two decades with modern spectrographs (e.g., VLT/MUSE,
93 Keck/KCWI, *JWST*/NIRSpec).
- 94 ▪ Proprietary Bayesian 3D methodologies: (?) introduced a 3D modeling technique designed
95 to recover lensed source kinematics. However, their underlying codebase remains private.
96 Additionally, their study was primarily a methodological demonstration using simulated
97 data; no application of their pipeline to real data was published.
- 98 ▪ Two-stage image-plane kinematic mapping: Alternative workflows attempt to bypass full
99 3D datacube forward-modeling by first extracting 2D kinematic maps (such as line-of-
100 sight velocity v_{los} and dispersion σ_v) in the image plane and subsequently ray-tracing
101 those 2D maps back to the source plane (?; ?). While useful, this decoupled approach
102 does not perform end-to-end forward-modeling. Extracting 2D kinematic markers directly
103 from distorted or overlapping lensed arcs introduces significant systematic errors due
104 to spatial blending, beam smearing, continuum contamination, and the destruction of
105 spaxel-level covariances. Furthermore, even this two-stage methodology lacks a publicly
106 available, standardized software package.

107 A fundamental design question during the development of IFUL was whether to contribute
108 3D modeling features directly upstream to *lenstronomy* or to create IFUL as an independent,
109 standalone package. *lenstronomy*'s architecture is tightly coupled to 2D image arrays via its
110 ImageModel and Data class abstractions. Modifying *lenstronomy*'s core codebase to natively
111 support 3D spectroscopic datacubes—which requires spaxel-by-spaxel spectral line synthesis,
112 3D velocity dispersion convolution kernels, continuum subtraction, source-plane kinematic
113 parameterizations, and customized 3D tensor likelihood evaluations—would require a structural
114 overhaul of *lenstronomy*'s data structures and API contract.

115 Instead, IFUL adopts an ecosystem wrapper design: it builds directly on top of *lenstronomy*,
116 using it “under-the-hood” as its underlying engine for 2D spatial ray-tracing, coordinate
117 transformations, and lens mass profile calculations (e.g., EPL, NFW, external shear). IFUL
118 extends this foundation by providing higher-level orchestrations for 3D datacubes, spectral
119 line synthesis, kinematic field mapping, and joint 3D likelihood computation. This design
120 choice delivers a specialized, modular, and user-friendly 3D modeling tool to the astronomical
121 community while avoiding code duplication and maintaining full interoperability with
122 *lenstronomy*'s mature 2D lensing engine.

123 Software design

124 Research impact statement

125 AI usage disclosure

126 Acknowledgements

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